

Does Playing at Home Give the Clippers an Advantage?

This season, the Los Angeles Clippers basketball team built a new stadium with a goal to create as big a home court advantage as possible. They built the largest dedicated fan section behind the away team's hoop in the first half. It has 51 uninterrupted rows of screened Clippers fanatics equipped with props to distract the players, designed uniquely for different opponents. Players have testified that this has a significant impact on their playing ability. We are curious to see if the efforts the Clippers organization has made have given them a statistically significant home court advantage, meaning a greater proportion of home games are won rather than lost.

Our retrospective study will include the population of the Los Angeles Clippers' regular-season schedule home games. We will pull data from our sampling frame of a list of all 41 Clippers regular-season games in the 2024-2025 season. There are two variables of interest: game location and game outcome. Both of these are categorical variables, as game locations are "home" or "away" and game outcomes are "win" or "loss".

Data was found through ESPN's database. It was cleaned to fit the needs of our study, only including whether a game was home or away and if the Clippers won or lost. After analyzing the data, it was found that the Clippers had a record of 30 wins and 11 losses at their home court.

ORIGINAL DATA:

[ESPN Clippers 2024-2025 Regular Season](#)

CLEANED DATA SET:

 Clippers 2024-2025 Season

One-Proportion Z-Test

$H_0: p = 0.5$

$H_A: p > 0.5$

Where p = the true proportion of Los Angeles Clippers 2024-2025 regular season home games won

$\alpha = 0.05$

In order to run the hypothesis test properly, three conditions will need to be checked. First, to check that each game is independent of the next, the Independence Assumption will pass with the Randomization Condition and the 10% Condition to determine if our sample is random and less than 10% of the population. Secondly, to check if our sample size is sufficient, the Sample Size Assumption will pass with the Success/Failure Condition to ensure we expect at least 10 successes (home court wins) and 10 failures (home court losses). Our null hypothesis states 50% of the Clippers' home games are won, while our alternate hypothesis states that more than 50% of the Clippers' home games are won. These hypotheses will be tested with an alpha level of 0.05, meaning that if the calculated p-value is above 0.05, we will fail to reject the null hypothesis, and if it is below 0.05, we will reject the null hypothesis.